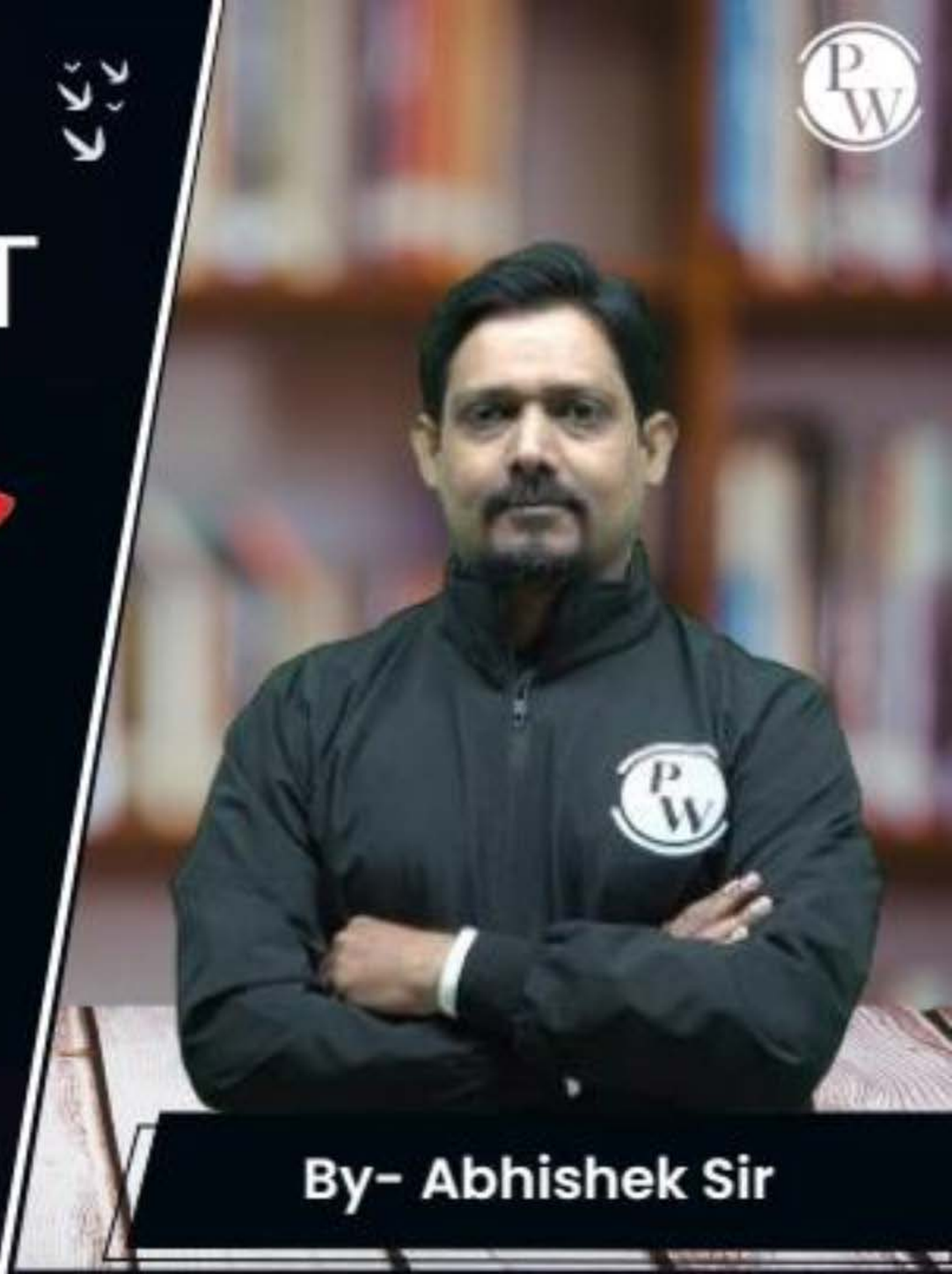
Computer Science & IT



C programming

Data Types & Operator

Lecture No. 04



By- Abhishek Sir

Recap of Previous Lecture



- topic
- topic
- topic
- topic
- topic

Arithmetic operator

precedence , Associativity

Slide

topics to be Covered



- topic
- topic
- topic
- topic
- topic

Compound statement

++, -- operator

Relational operator

Slide



```
a+=b;   =  a=a+b
a*=10;  =  a=a*10
a/=2;   ,  a=a/2;
a%=3;      a=a%3;
a-=c;      a=a-c;
```

assignment & Arthmetic operators.

Slide



Toipc: ++ and --

```
int a = 10;
printf("%d", a+10);
printf("%d", a);
```

No assignment No updation

code segment

a does not modify.

Slide



Toipc: ++ and --

++, -- Modifiable operator

++, -- Increment/ Decrement by 1

++, -- operator works with only variable

Slide



Toipc: ++ and --

int a = 10

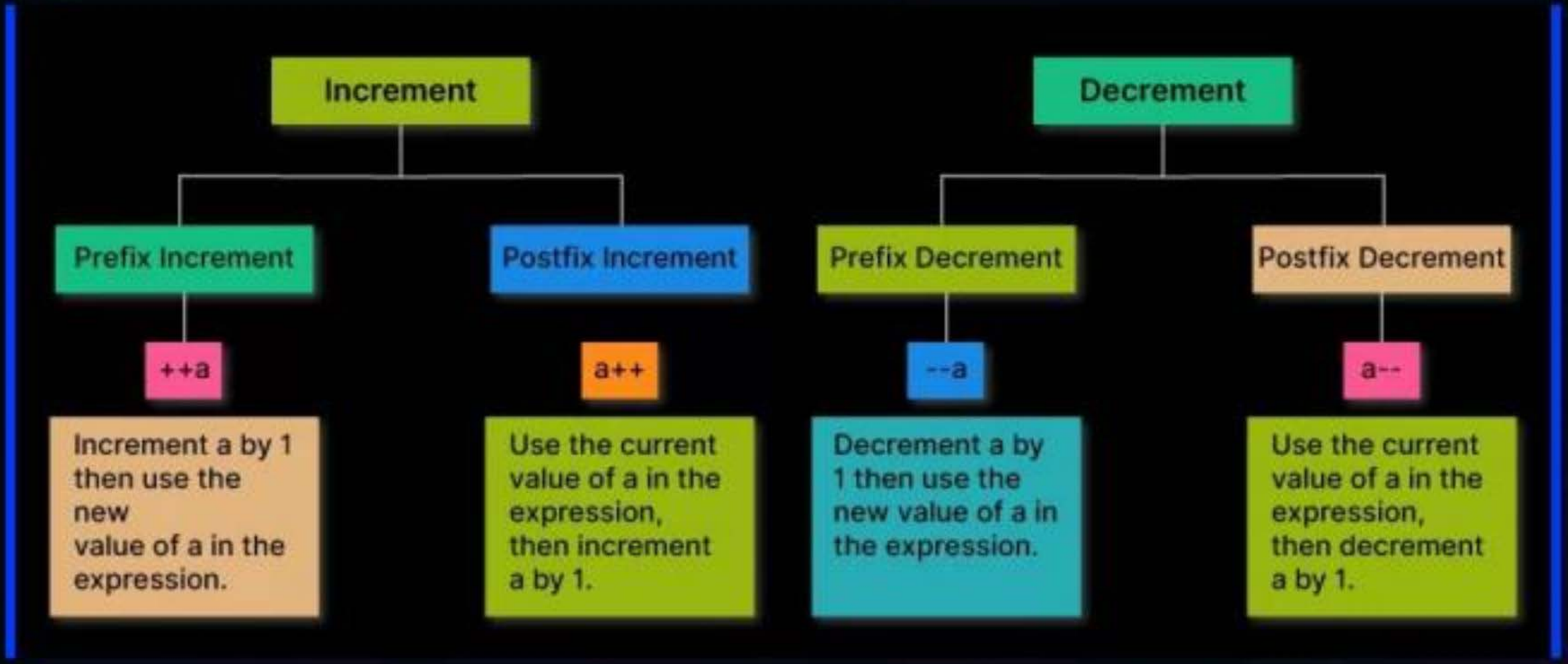
{ a++; — post increment
  ++a; — pre increment } a = a + 1

a--; — post decrement
--a; — pre - decrement } a = a - 1

Slide



Toipc: Increment & decrement Operator



Slide



Toipc: Question

```
#include <stdio.h>
      int main ()  {
       int x=5;
       int y;
       y = ++x;
      printf("%d %d", x, y);
     }
```

x 5 → poe_Increment

x 6

y = x · y 6

- first Increment then use in expression

The output of the program

(A) 5,5

(B) 5,6

(C) 6,5

(D) 6,6

Slide



Toipc: Question

```
#include <stdio.h>
      int main ()   {
       int x=5;
       int y;
       y = ++x;
      printf("%d %d", x, y);
     }
```

The output of the program

(A) 5,5 (B) 5,6 (

C) 6,5 (D) 6,6

Answer

D

Slide

Toipc: ++ and --

```c
#include <stdio.h>

int main(){
    int x = 5, y, b=10;

    y  = x++;
    y = ++x;
    y = x++*b;
    y = ++x*b;
    y  = x--;
    y = --x;
    y = x--*b;
    y = --x*b;

    printf("%d\n",x);
    printf("%d",y);

    return 0;
}
```

x: 5 6 7 8 9 8 7 6 5

y = x++; → y = 5

y = ++x; → y = 7

y = x++*b; → y = 7*10 = 70

y = ++x*b; → y = 9*10 = 90

y = x--; → y = y

y = --x; → y = 7

y = x--*b; → y = 7*10 = 70

y = --x*b; → y = 5*10 = 50

printf("%d\n",x); → 5

printf("%d",y); → 50

x+y = 55

Slide



Toipc: Question

What is the output of the program?

```
#include <stdio.h>
      int main ()  {
        int x=5;
        int y;
        y = x++;
         printf("%d %d", x, y);
     }
```

x 5 → 6

y = x++ y 5

6 5

first use in expression then Increment the value

post Increment

Output of the program is_____

(A) 5,5

(B) 5,6

(C) 6,5

(D) 6,6

Slide



Toipc: Question

What is the output of the program?

```
#include <stdio.h>
      int main ()   {
        int x=5;
        int y;
        y = x++;
         printf("%d %d", x, y);
     }
```

Output of the program is_____

(A) 5,5

(B) 5,6

(C) 6,5

(D) 6,6

Slide



topic : variable

Consider the following program:

```
#include <stdio.h>
int main()
{
char ch=119;
ch=ch+6;
printf("%d", ch);
return 0;
}
```

The output is- _____.

Slide

Toipc: GATE 2017

Consider the following C program.

```
#include<stdio.h>
int main () {
    int m=10;
    int n, n1;
    n=++m;
    n1=m++;
    n--;
    --n1;
    n-=n1;
    printf("%d", n);
    return 0;
}
```

The output of the program is [0]

m | 10 | 11 | 12

n = 11 | 10

n_1 = 11 | 10

n-- ; --n1 ; → n-=n1 ⇒ n = n - n_1

10 - 10

n = 0

Slide



Char

```
#include <stdio.h>
int main() {
        char ch = 65+7;
        printf("%c", ch);
        return 0;
}
```

(a)A

(B)N

(c)n

(d) z

Slide



Toipc:Question

Dividend = quotient * divisor + Remainder

```
#include<stdio.h>
   int main(){
   int x;
   x=  -14%8+2%-8;
   printf("%d",x);
   return 0 ;
   }
```

-14%8

$\frac{-14}{8}$ Q = -1

R = -6

-6 + 2

= -4

The value of x is

(a) -6
(b) 7
(c) -4
(d) 9

$\frac{2}{-8}$ = q = 0

r = 2

Slide



Character

(d)
Unsigned value for –134= 256 – 134 = 122.
Hence, ‘z’ is printed.



Character

Consider the following program:

```
#include <stdio.h>
int main() {
char ch=119;
ch=ch+6;
printf("%d", ch);
return 0;
}
```

The output is- _____.



++ operator not applied on

as well as -- operator Not allowed

Constant · 5++; ⟵ Error - L-value required

expression (2+3%-8)--

2+3 =5

Modification

pointer & Memory location

Slide



topic: Relational Operator

Relational operators

> < >= <= == !=

Slide



topic: Relational Operator

<	>	>=	<=	==	!=
Less than	Greater than	Greater than equal to	Less than equal to	Exactly equal	Not equal to

5>6

Slide



topic: Relational Operator Example

```
#include<stdio.h>
int x = 40;
int  main() {
        printf("%d\n", 30>40);
        printf("%d\n", 30>=40);
        printf("%d\n", 30==40);
        printf("%d\n", 30!=40);
        printf("%d\n", 40!=30);
        printf("%d\n", 40==40);
        printf("%d\n", 50>50);
        printf("%d\n", 50<=50);
        return 0 ;
}
```

0
0
0
1
1
1
0
1

0 or 1

false true

30 != 30

false 0

Slide



topic: Relational Operator Precedence

3	* / %	Multiplication, division, and modulus	left to right
4	+ -	Addition and subtraction	left to right
6	< <= > >=	Relational less than and less than or equal to Relational greater than and greater than or equal to	left to right
7	== !=	Relational equal to and not equal to	left to right

$5 > 6 == 1$

$0 == 1 == 0$

Slide



topic: Relational Operator Example

```
#include<stdio.h>
int  main() {
    int x = 40, y = 50 , z = 30;
      printf("%d\n", x>y<z);

}
```

The value printed by the program

is ____

```
int a=4, b=6, c=20, d=30
printf("%d", a+b+c>=d != b-a);
```

4+6+20>=30 != 6-4

30>=30 != 2

1 != 2 = 1



topic: Relational Operator Example

```
#include<stdio.h>
int  main() {
    int x = 40, y = 50 , z = 30;
      printf("%d\n", x>y<z);
}
```

The value printed by the program is 1

(40>50)<30 Left to right

0<30 = 1

Slide



topic: Relational Operator Example

```
#include<stdio.h>
int main(){
        int a;
        a=52>54>33>30>28>19>20;
        printf("%d",a);
        return 0;
}
```

The value printed by the program is 0

52>54

0>33

0>30

0>28

0>19

0>20

0

```
#include<stdio.h>
int main() {

    int num1 = 5; // 6 // 0
    int num2 = 3; // 4
    int num3 = 2; // 1

    num1 = num2++ * --num3 / ++num1;
    printf("%d, %d, %d", num1, num2, num3);
    return 0;
}
```

value pointed

(A) 0, 4, 1 ✓

(B) 6, 4, 1

(C) 6, 3, 2

(D) 0, 3, 2,

Last class

left Associativity

3 * 1 / 6 = 3/6 = 0



2 mins Summary

Topic — Increement & decreement

Topic — ++ . − − , pust poe

Topic — % operator

Topic — Relational operator

Topic

Slide

THANK - YOU